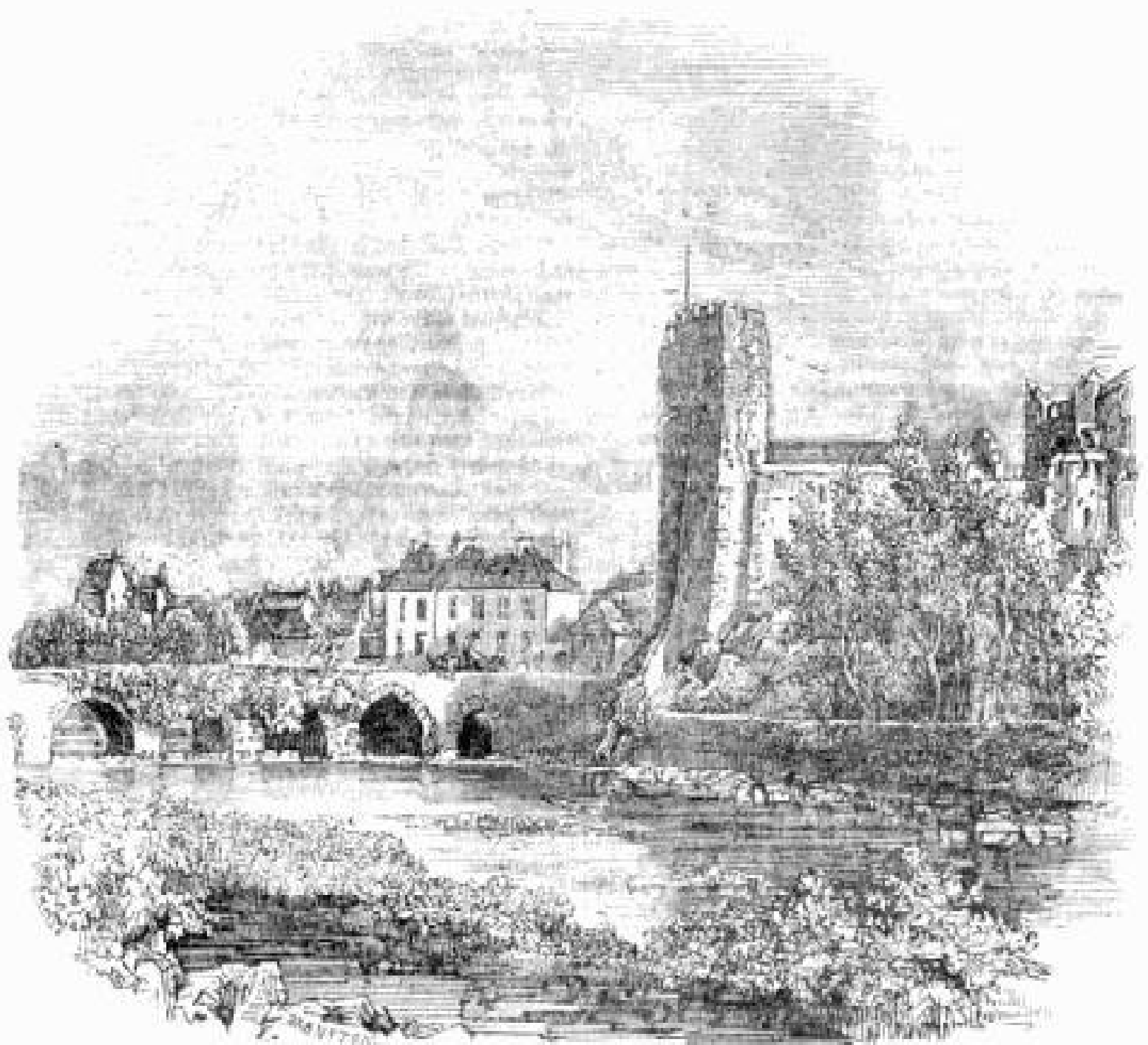


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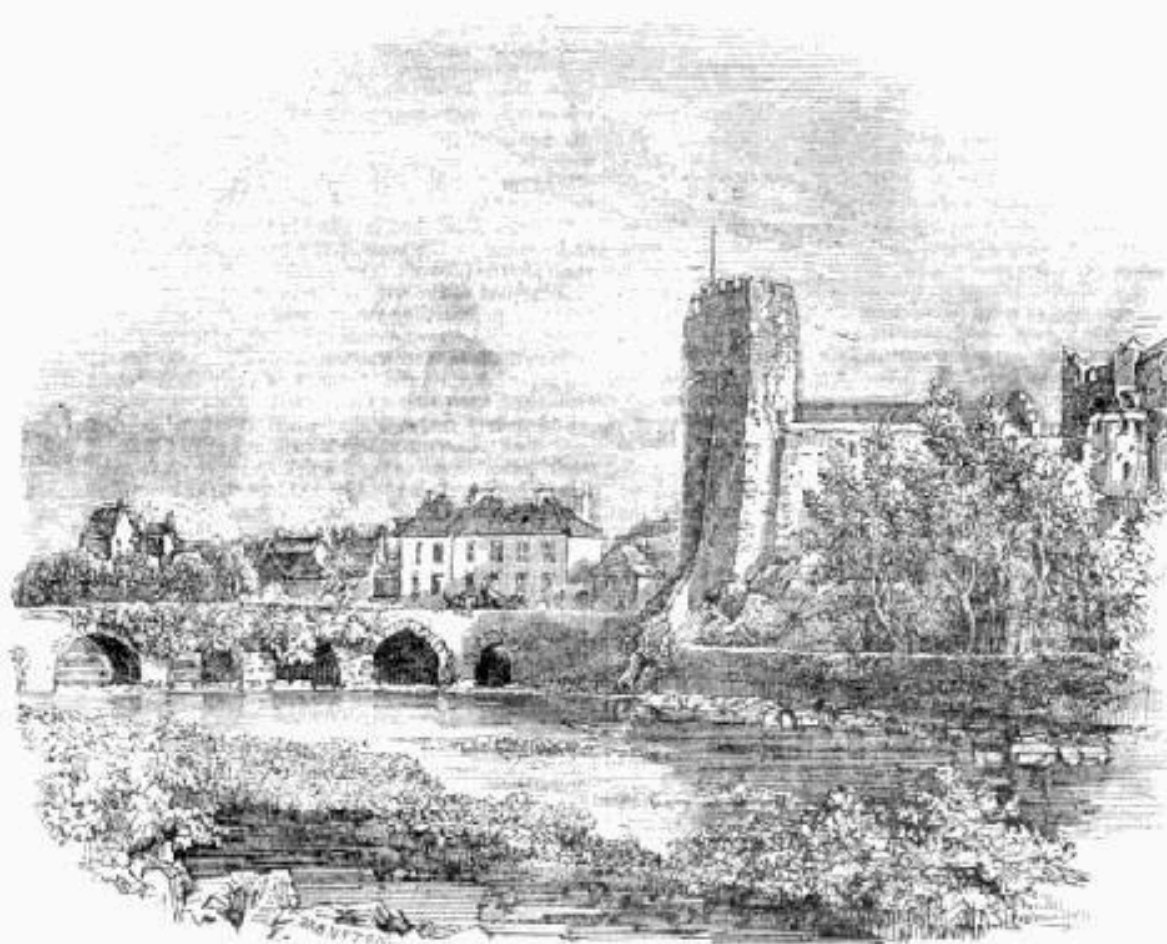
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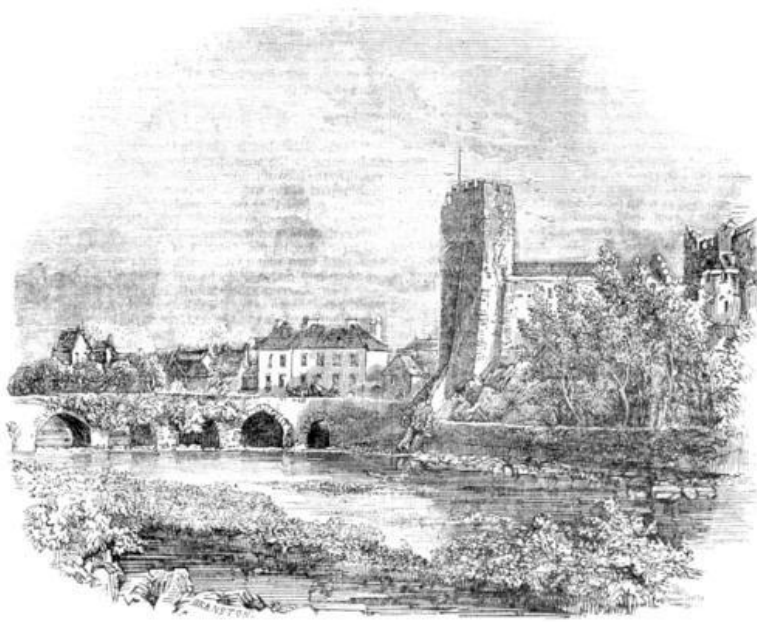
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CAHIR CASTLE, COUNTY OF TIPPERARY

To a large portion of our readers it will be scarcely necessary to state, that the little town of Cahir is in many respects the most interesting of its size to be found in the province of Munster, we had almost said in all Ireland; and that, though this interest is to a considerable extent derived from the extreme beauty of its situation and surrounding scenery, it is in an equal degree attributable to a rarer quality in our small towns—the beauty of its public edifices, and the appearance of neatness, cleanliness, and comfort, which pervades it generally, and indicates the fostering protection of the noble family to whom it belongs, and to whom it anciently gave title. Most of our small towns require brilliant sunshine to give them even a semi-cheerful aspect: Cahir looks pleasant even on one of our characteristic gloomy days. As it is not, however, our present purpose to enter on any detailed account of the town itself, but to confine our notice to one of its most attractive features—its ancient castle—we shall only state that Cahir is a market and post town, in the barony of Iffa and Offa West, county of Tipperary, and is situated on the river Suir, at the junction of the mail-coach roads leading respectively from Waterford to Limerick, and from Cork by way of Cashel to Dublin. It is about eight miles W.N.W. from Clonmel, and the same distance S.W. from Cashel, and contains about 3500 inhabitants.

The ancient and proper name of this town is *Cahir-duna-iascaigh*, or, the circular stone fortress of the fish-abounding Dun, or fort; a name which appears to be tautological, and which can only be accounted for by the supposition that an earthen *Dun*, or fort, had originally occupied the site on which a *Cahir*, or stone fort, was erected subsequently. Examples of names formed in this way, of words having nearly synonymous meanings, are very numerous in Ireland, as *Caislean-dun-more*, the castle of the great fort, and as the Irish name of Cahir Castle itself, which, after the erection of the present building, was called *Caislean-na-caherach-duna-iascaigh*, an appellation in which three distinct Irish names for military works of different classes and ages are combined.

Be this, however, as it may, it is certain that a *Cahir* or stone fort occupied the site of the present castle in the most remote historic times, as it is mentioned in the oldest books of the Brehon laws; and the Book of Lecan records its destruction by Cuirreach, the brother-in-law of Felemy Rechtmar, or the Lawgiver, as early as the third century, at which time it is stated to have been the residence of a female named Badamar. Whether this *Cahir* was subsequently rebuilt or not, does not appear in our histories as far as we have found; nor have we been able to discover in any ancient document a record of the erection of the present castle. It is stated indeed by Archdall, and from him again by all subsequent Irish topographers, that Cahir Castle was erected prior to the year 1142 by Conor-na-Catharach O'Brien, king of Thomond. But this is altogether an error. No castle properly so called of this class was erected in Ireland till a later period, and the assertion of Conor's having built a castle at Cahir is a mere assumption drawn from the cognomen *na-Catharach*, or of the Cahir or Fort by which he was known, and which we know from historical evidences was derived not from this Cahir on the Suir, but from a Cahir which he built on an island in Lough Derg, near Killaloe, and which still retains his name. The true name of the founder of Cahir Castle, and date of its erection, must therefore remain undecided till some record is found which will determine them; and in the meantime we can only indulge in conjecture as to one or the other. That it owes its origin, indeed, to some one of the original Anglo-Norman settlers in Ireland, there can be little doubt, and its high antiquity seems unquestionable. As early as the fourteenth century, it appears to have been the residence of James *Galdie* (or the Anglified) Butler, son of

James, the third Earl of Ormond, by Catherine, daughter of Gerald, Earl of Desmond—whose descendant Thomas Butler, ancestor to the present Earl of Glengal, was advanced to the peerage by letters patent, dated at Dublin the 10th November 1543 (34 Henry VIII.) by the title of Baron of Cahir.

In the subsequent reigns of Elizabeth and the unfortunate Charles I, Cahir Castle appears as a frequent and important scene in the melancholy dramas of which Ireland was the stage, and its history becomes a portion not only of that of our country generally, but even in some degree of that of England.

It will be remembered, that, when by the battle of the Blackwater in 1598 the English power in Ireland was reduced to the lowest state, and the queen felt it necessary to send Robert Devereux, Earl of Essex, with an army of more than 20,000 men—the largest body, as the Four Masters state, that had ever before come into Ireland since the time of Strongbow—to subdue the rebels, that unfortunate favourite, neglectful of the instructions imperatively given to him that he should prosecute the Ulster rebels, and plant their strongholds with garrisons, marched into Munster, where the only deed of importance he achieved was the taking of Cahir Castle, and the forcing of the Lord Cahir and some other disaffected noblemen of Munster to submit, and accept the queen's protection. The only favourable result of this misguided enterprise, as Morrison acquaints us, "was the making a great prey of the rebels' cattle in those parts; he cast the terror of his forces on the weakest enemies, whom he scattered and constrained to fly into woods and mountains to hide themselves." But these weak rebels did not remain long inactive, or exhibit weakness in attack; and the earl's journey back to Dublin towards the end of July was marked by a series of disasters that sealed his doom; or, as the Four Masters remark, "The Irish afterwards were wont to say that it were better for the Earl of Essex that he had not undertaken this expedition from Dublin to Hy-Conell Gaura, as he had to return back from his enterprise without receiving submission or respect from the Geraldines, and without having achieved any exploit except the taking of Cahir-duna-iasgach."

The taking of Cahir Castle was not effected without considerable trouble, though it is stated that Essex's army amounted to 7000 foot and 1300 horse. O'Sullivan states that the siege was prolonged for ten days, in consequence of the Earl of Desmond and Redmond Burke having come to its relief; and the Four Masters state in their Annals that "the efforts of the earl and his army in taking it were fruitless, until they sent for heavy ordnance to Waterford, by which they broke down the nearest side of the fortress, after which the castle had to be surrendered to the Earl of Essex and the queen." This event occurred on the 30th of May 1599.

As Morrison, however, remarks, the submission of the Lord Cahir, Lord Roche, and others, which followed on this exploit, were only feigned, as subsequent events proved. After the earl's departure, they either openly joined the rebel party again, or secretly combined with them; and on the 23d of May in the year following, the Castle of Cahir was surprised and taken by the Lord Cahir's brother, and, as it was said, with his connivance. Of this fact the following account is given by Sir George Carew in his *Pacata Hibernia*:—

"The president being at Youghall in his journey to Cork, sent Sir John Dowdall (an ancient captain in Ireland) to Cahir Castle, as well to see the same provided of a sufficient ward out of Captain George Blunt's company, as to take order for the furnishing of them with victuall, munition, and other warlike provision; there he left the eighth or ninth of May, a serjeant, with nine-and-twenty soldiers, and all necessary provision for two months, who notwithstanding, upon the three-and-twentieth of the same, were surprised by James Galdie, alias Butler, brother to the lord of Cahir, and, as it was suspected by many pregnant presumptions, not without the consent and working of

the lord himself, which in after-times proved to be true. The careless security of the warders, together with the treachery of an Irishman who was placed sentinel upon the top of the castle, were the causes of this surprise.

“James Galdie had no more in his company than sixty men, and coming to the wall of the bawne of the castle undiscovered, by the help of ladders, and some masons that brake holes in some part of the wall where it was weak, got in and entered the hall before they were perceived. The serjeant, named Thomas Quayle, which had the charge of the castle, made some little resistance, and was wounded. Three of the warde were slaine; the rest upon promise of their lives rendered their armes and were sent to Clonmell. Of this surprise the lord president had notice when he was at Kilmallock, whereupon he sent directions for their imprisonment in Clonmell until he might have leisure to try the delinquents by a marshals’ court. Upon the fourth day following, James Butler, who took the castle, wrote a large letter to the president, to excuse himself of his traitorly act, wherein there were not so many lines as lies, and written by the underhand working of the lord of Cahir his brother, they conceiving it to be the next way to have the castle restored to the baron.”

Cahir Castle was, however, restored to the government in a few months after, as detailed in the following characteristic manner by Sir George Carew:—

“Towards the latter end of this month of August, the lord deputy writing to the president about some other occasions, it pleased him to remember Cahir Castle (which was lost as before you have heard), signifying that he much desired to have that castle recovered from the rebels, the rather because the great ordnance, or cannon, and a culverin being left there by the Earl of Essex, were now possessed by the rebels. This item from the lord deputy spurred on the president without further delay to take order therein, and therefore presently by his letters sent for the lord of Cahir to repair unto him, who (as before you have heard) was vehemently suspected to have some hand both in the taking and keeping thereof. The Baron of Cahir being come, the council persuaded him to deal with James Butler (nicknamed James Galdie) his brother, about the redelivering thereof to her Majesty’s use; but his answer was, that so little interest had he in his brother, as the meanest follower in all his country might prevail more with him than himself (for he was unwilling to have the castle regained by the state, except it might again be left wholly to him, as it was before the first winning thereof); which the president surmising, told him, that if it might speedily be yielded up unto him, he would become an humble suitor to the lord deputy (in his behalf) for the repossessing thereof; otherwise he would presently march with his whole army into those parts, and taking the same by force, he would ruin and rase it to the very foundation, and this he bound with no small protestations. Hereupon Justice Comerford being dispatched away with the lord of Cahir, they prevailed so far with young Butler, that the castle, upon the twenty-ninth following, was delivered to the state, as also all the munitions, and the great ordnance conveyed to Clonmell, and from thence to Waterford.”

Notwithstanding these imputed crimes of the Lord Cahir, and the open treason of his brother, he received the queen’s pardon by patent, dated the 27th day of May 1601, and died in possession of his castle and estates in January 1628. His brother James Galdie, however, lived to take his share in the troubles that followed in 1641, and suffered accordingly.

From these stories of violence and treachery we turn with pleasure to record a fact of a peaceful character, in which Cahir Castle appears as a scene of hospitality and splendid revelry. This occurred in 1626, when the Lord Deputy Falkland, in making a tour of Ireland, after residing a considerable time at the Earl of Ormond’s castle at Carrick-on-Suir, in some time after came to the lord of Cahir, and was entertained by him in his castle with the greatest splendour.

But if these old walls had tongues, they could probably tell us of many scenes of a different character from that we have just narrated, and of which one has been dimly preserved in history. Immediately after the death of Thomas, the fourth Lord Cahir, in 1628, as already stated, his property having passed to his only daughter and heir Margaret, who was married to her kinsman Edmund Butler, the fourth Lord Dunboyne, the latter, while residing in this castle with his wife, slew in it, or murdered, perhaps, would be the more correct word, Mr James Prendergast, the owner of Newcastle, for which he was confined a prisoner in the Castle of Dublin; and his Majesty having granted a commission on the 4th of June in that year, constituting the Lord Aungier high steward of Ireland for the trial of his lordship, he was tried by his peers accordingly, but acquitted, fifteen peers voting him innocent, and one, the celebrated Lord Dockwra, voting him guilty.

During the troubles which followed on the rebellion of 1641, Cahir Castle was taken for the Parliament, by surrender, in the beginning of August 1647 by Lord Inchiquin; and it was again taken in February 1650 by Cromwell himself, the garrison receiving honourable conditions. The reputation which the castle had at this period as a place of strength will appear from the account of its surrender as given in the manuscripts of Mr Cliffe, secretary to General Ireton, published by Borlase. After observing that Cromwell did not deem it prudent to attempt the taking of Clonmel till towards summer, he adds, that he “drew his army before a very considerable castle, called Cahir Castle, not very far from Clonmel, a place then possessed by one Captain Mathews, who was but a little before married to the Lady Cahir, and had in it a considerable number of men to defend it; the general drew his men before it, and for the better terror in the business, brought some cannon with him likewise, there being a great report of the strength of the place, and a story told the general, that the Earl of Essex, in Queen Elizabeth’s time, lay seven or eight weeks before it, and could not take it. He was notwithstanding then resolved to attempt the taking of it, and in order thereunto sent them this thundering summons:—

‘SIR—Having brought the army and my cannon near this place, according to my usual manner in summoning places, I thought fit to offer you terms honourable for soldiers, that you may march away with your baggage, arms, and colours, free from injuries or violence; but if I be, notwithstanding, necessitated to bond my cannon upon you, *you must expect what is usual in such cases*. To avoid blood, this is offered to you by

Your servant,

O. CROMWELL.

For the Governor at Cahir Castle,
24th February 1649’ (1650.)

“Notwithstanding the strength of the place, and the unseasonableness of the time of the year, this summons struck such a terror in the garrison, that the same day the governor, Captain Mathews, immediately came to the general and agreed for the surrender,”—&c.

It was well for Captain George Mathews, or Mathew, as the name is now generally written, and his garrison too, that he had not the hot-headedness of an Irishman, or he would probably have set this “thundering summons” at defiance, and Cahir Castle would not only have shared the fate of most Irish fortresses at that period, but, what would have been a far greater loss, the Apostle of Temperance, who has done as much to regenerate the people of Ireland as Cromwell did to destroy them, would in all human probability never have existed.

But we are exceeding the limits assigned to us, and can only add a few words of general description. Cahir Castle is built upon a low rugged island of limestone, which divides the water of

the Suir, and which is connected by a bridge with the two banks of the river. It is of considerable extent, but irregular outline, consequent upon its adaptation to the form and broken surface of its insular site, and consists of a great square keep, surrounded by extensive outworks, forming an outer and an inner ballium, with a small court-yard between the two; these outworks being flanked by seven towers, four of which are circular, and three of larger size, square. From a very interesting and accurate bird's-eye view of the castle, as besieged by the Earl of Essex, given in the *Pacata Hibernia*, we find, that notwithstanding its great age, and all the vicissitudes and storms it has suffered, it still presents, very nearly, the same appearance as it did at that period; and from the praiseworthy care in its preservation of its present lord, it is likely to endure as a beautiful historical monument for ages longer.

P.

IRISH MUSICIANS OF THE LAST CENTURY, STORY OF DOCTOR COGAN.

In this grave cigar-smoking age of ours, in which Irishmen exhibit so little of the love of fun and merriment—the drolleries and *escapades* which distinguished them in preceding ages—it is a pleasant thing to us septuagenarians to look back occasionally to our youthful days, and call up from the storehouse of our memories the merry men whom and whose merry freaks we were either familiar with, or at least had heard of or seen. One of these choice spirits is just now present with us in our mind's eye, and we are certain that we have only to mention his name, to bring him equally before a great number of our Dublin readers. We mean the late musical doctor, John Cogan. There, now, Dublin readers, some thousands of you at least have the man before you, though many of you are unfortunately too young to have heard his exquisitely delicate and expressive hands on the piano, extemporising with matchless felicity upon Garryowen or some other melody of Old Ireland; or participated in his playful and always inoffensive merriment and good humour. Even the youngest of you, however, must surely remember the little man—little indeed in size, but every inch of him a gentleman, who but a few years since might be occasionally seen taking an airing, when the sun shone on him, in Sackville Street, sometimes leaning on his servant's arm, and at others driven in his pony-phaeton, which his prudence in youth had enabled him to secure for his days of feebleness and old age. That pleasant intellectual countenance, bright and playful as his own music even to the last, has disappeared from amongst us; but the memory of such a man should not be allowed to die, and we will therefore, while in the vein, devote a column of our Journal to a sketch of one of the many incidents remembered of his long life, as illustrative in some degree not only of his character, but also of that of society in Dublin during the last century.

From what we have already stated, it will have appeared that Doctor Cogan was not only great as a musical performer, but also as a performer in innocent waggery. It would indeed have been difficult to determine in which performance he most excelled, or whether he most loved his music or his joke. He was not only a good theorist, but loved a bit of *harmony* intensely, and a *laughing chorus* was his prime delight. Those he would often accompany or direct as occasion required, to heighten the pleasures of a musical treat, when he rarely neglected a happy opportunity of introducing some *vivace* movement of his own composing, provided he could previously prepare a *score* of good fellows capable of performing effectively the several parts assigned them in it, which among his apt compeers was rarely a difficult task. A lover of good cheer and hospitality, which he both gave as well as partook of with a true Irish spirit, it was a settled point with the Doctor that brother professors should at all times live in harmony with each other, and receive brotherly encouragement; nor were such feelings of an exclusively national character, but extended equally to foreigners coming to Ireland, who, if at all known to fame, were sure of receiving a friendly and *cead mile failte* reception at his hands. If, it is true, he could on such occasions indulge a little innocent joke, by playing off a specimen of Irish *counterpoint* at the expense of such visitors, it was so much the more agreeable to him, as in the following instance of the

concerted movement which he got up to do honour to the celebrated violinist Pinto, who visited our city about sixty years since. But before we detail the circumstances attendant on this reception, it is necessary that we should tell our worthy readers something of the person who was selected by the Doctor to play a leading part—the principal fiddle—on the occasion; and the more particularly as his name is unknown to the great majority of the present generation, and almost forgotten by the few who may still survive him.

The person we allude to was Robert Meekins, or, as he was familiarly called, “Bob,” a violinist of great tavern-playing notoriety in his day. Like his brother professors, the harpers of the last century, of whom Mr Bunting has given us such characteristic anecdotes, Bob was a thoroughly Irish musician in every sense of the word; and though, as we believe, he had never travelled out of Dublin, his native city, few were found to equal him on his instrument either in tone, execution, or expression of feeling. From the earliest period of his musical studies, however, he had indulged in a wild and extemporaneous mode of practice, which proved most injurious to his professional career in after life, and unfortunately for him, being moreover an inveterate hater of *dry* study, Bob more frequently wetted his whistle than he rosined his bow. Under the influence of such *bad practice* he became at last incurably vicious, and rarely kept within reasonable bounds, either in the way of drinking or fiddle-playing. Indeed, whatever command poor Bob retained over his instrument, he had none over himself. Leader after leader sought to curb him in his wild extravagances of style, in the vain hope of diverting his great natural musical powers into legitimate courses; but Bob would never be led, and as to driving him, that was found to be equally impracticable. He would go his own way, and no other. He would read concerted music, not as it was intended, but as he thought it should be. His passion for *obligatoes* was unconquerable, and he rarely arrived at an *ad libitum* that he did not avail himself of it with a vengeance; and thus, while his brother musicians were attending to the pauses, perfectly content with the single note before them, an impromptu cadence would be heard meandering through a chord, telling of Bob’s wanderings, and he the while so absorbed as to be equally heedless of the elbow-punchings of his neighbours, the authority of his leader, or the intentions of the composer. No composer indeed came up to his fancy—entirely; something was always wanting, and his fingers were ever upon the alert to supply that something which was not set down for him: and should a remonstrance come from the leader, it but too frequently produced a *presto* movement on the part of Bob, leaving a vacancy in the orchestra to be filled up as it might, at the shortest possible notice. Vain of his powers, and scorning restraint, his kicks against orchestral rule became beyond all bearing, and so he was himself at last kicked out from all decent musical society. Thus finding himself alone, he naturally turned *solo* player, and became one of the lions of Dublin, drawing nightly crowds to the taverns he frequented, where he could indulge his love for flights of fancy to his heart’s content. But, unfortunately for him, in this new sphere he was enabled by the liberal contributions of his admirers to indulge also without restraint that more fatal passion for drink which had proved his bane through life, leading him step by step, as usual with such reckless characters, to an untimely and degraded grave. It is generally believed that poor Bob Meekins died from the effects of intemperance in some wretched doorway in an alley of our city.

Such, then, was the person selected by Doctor Cogan to perform a principal part in the little musical drama which he had prepared for the reception of the great foreign violinist of the day, and the place chosen for its performance was the once celebrated hotel or tavern called the Pigeon-house, which at that period was the common resort for the meetings or departures of friends to or from England by the Holyhead packets. Thither accordingly the Doctor and his musical companions repaired, to await the expected arrival of the Signor, and ordered dinner with the determination that he should be their guest. It is not necessary to dilate upon the reception given to

the brother professor, or to particularise all the good things that were said, sung, and eaten upon the occasion. It is sufficient to say that every thing passed off in true Hibernian style, to the astonishment as well as gratification of Pinto, who was delighted to find himself surrounded by so many new and warm-hearted friends, each keeping up the tide of merriment by a rapid circulation of the bottle amid the joyous flow of song, jest, and laugh. But where was Bob all this time? He was placed in an adjoining passage awaiting a silent signal, and being primed for action, was impatient for the moment of attack upon the excitable nerves of the delighted Italian. This signal was at length given, and so effectually arranged were the parts given to each of the Doctor's apt pupils, that as the soul-thrilling tones of Bob's violin vibrated through the room, it seemed to produce no other effect upon their ears than a *sotto voce* expression of displeasure, or *forzando* of horror. All this seemed quite spontaneous, and was at the same time so judiciously managed as to allow the instrument to predominate over the voices, and thus enable the practised ear of Pinto to discover in the invisible minstrel a master spirit—nor did the well-timed *crescendo* of "Turn the scraping villain out," "Curse the noisy blackguard," &c. &c. arrive at its climax, until Bob's varied and expressive execution had completely bewildered the poor Signor with amazement. To him, indeed, the scene was one as unusual as it was unexpected; and when silence was somewhat restored, he eagerly asked in his broken English whence the tones had come; and truly ludicrous were the varied expressions of the Italian's intellectual countenance on being assured by the Doctor and his assistants that the performer who had so enraptured him was a rascally itinerant fiddler, who gained a precarious livelihood by scraping at taverns. The effect may easily be imagined. The Signor insisted upon seeing him; and when Bob's whisky face and tattered habiliments became visible, Pinto sat fixed in mute bewilderment, conjuring up in his excited imagination the apparition of a Meekins at the corner of every street; and the success of the Doctor's joke was complete, when the poor Italian, with a forlorn and chopfallen visage, was heard to mutter, "Lit-el fid-el—lit-el fid-el—you call—if dis lit-el fidel, me go back, me no use!"

A simultaneous burst of laughter was the response to these hurried and broken accents of surprise and chagrin. But enough was effected, and in quick compassion for poor Pinto's feelings, he was at once made to understand the whole contrivance, on which he laughed as loudly as any of the merry Irish group around him. The scene of joyousness was kept up till an *early* hour, during which Meekins occasionally revelled in the music of his own dear land, to the increased delight not only of the Signor, but of all present on the occasion.

W.

THE INQUIRY.

Tell me, ye winged winds,
That round my pathway roar,
Do ye not know some spot
Where mortals weep no more?
Some lone and pleasant dell,
Some valley in the west,
Where, free from toil and pain,
The weary soul may rest?
The loud wind dwindled to a whisper low,
And sigh'd for pity as it answered "No!"

Tell me, thou mighty deep,
Whose billows round me play,
Knowest thou some favour'd spot,
Some island far away,
Where weary man may find
The bliss for which he sighs?
Where sorrow never lives,
And friendship never dies?
The loud waves, rolling in perpetual flow,
Stopp'd for a while, and sigh'd, to answer. "No!"

And thou, serenest moon,
That, with such holy face,
Dost look upon the earth
Asleep in night's embrace,
Tell me, in all thy round
Hast thou not seen some spot,
Where miserable man
Might find a happier lot?
Behind a cloud, the moon withdrew in woe,
And a voice, sweet, but sad, responded "No!"

Tell me, my secret soul,
O! tell me, Hope and Faith,
Is there no resting-place
From sorrow, sin, and death?
Is there no happy spot
Where mortals may be bless'd—
Where grief may find a balm,
And weariness a rest?
Faith, Hope, and Love, best boons to mortals given,
Waved their bright wings, and whisper'd, "Yes! in Heaven!"

—Mackay's Poems

ON THE SUBJUGATION OF ANIMALS BY MEANS OF CHARMS, INCANTATIONS, AND DRUGS.

Second Article.

SERPENT-CHARMING AS PRACTISED BY THE JUGGLERS OF ASIA.

In my last paper I endeavoured to furnish my readers with a description of serpent-charming, as at present practised by the jugglers of Egypt, Arabia, and India. I now come to a review of the opinions maintained respecting this mysterious art, and the secret on which it depends, by some of the most eminent philosophers who have turned their attention to the subject.

These opinions are as various as they are numerous, no two individuals who have written upon the practice agreeing in any one particular, save only their determination to regard the whole affair as an imposture—the snake-charmers as clever and designing cheats, and all who believed in the reality of their performances, as silly dupes. I shall merely advert to some of the most striking of these suppositions, and then proceed to an investigation of their merits, ere advancing my own theory on the subject.

Many travellers who have written on the practice of serpent-charming have declared it as their conviction that the process is based in deception, that is, that the serpents charmed forth from holes are by no means wild creatures, who really and naturally inhabit those recesses, but animals which have been previously tamed, their poisonous fangs extracted, and placed there by the juggler or an accomplice, in order to the performance of his pretended miracle. Amongst the most prominent of these objectors are to be found the Abbé Dubois and the traveller Denon; and the latter author even goes so far as to affirm that the secret of the Psylli was a piece of nonsense that he might easily have discovered had he been so disposed. A precious traveller truly! to have had it in his power to discover a secret that a hundred naturalists would have given their very eyes to become acquainted with, and yet to neglect taking the necessary trouble. Ah, Monsieur Denon, how you do remind me of the witty fable of the fox and the sour grapes! The Abbé Dubois, though equally sceptical, does not venture to handle this mysterious subject quite so cavalierly as Denon. He says that the Psylli perform various *tricks* with serpents, which, though apparently terrible, are not very dangerous, as they *always* take the precaution to have the fangs previously removed, and to have with them the venomous vesicle extracted. He likewise informs us that they are *supposed* to have the power of charming those dangerous reptiles, and of commanding them to approach and surrender themselves at the sound of music; and he quotes the passages of scripture to which I referred in my preceding article, as confirmatory of the authenticity of the practice; yet he will not admit that even this mass of evidence will convince that the charmer's art is aught but an imposture. "Without dwelling," says he, "on the literal accuracy of this striking passage of Holy Scripture, I may confidently affirm that the skill which the Indian *pretenders to enchantment* claim in this particular, is rank imposture. The trick consists in placing a snake, previously tamed and accustomed to music, in some remote place, and they manage it so that in appearing accidentally to approach that place, and beginning to play, the snake comes forward at the wonted sounds.

When they enter into an agreement with any simpleton who fancies that his house is infested with serpents—a notion which they sometimes contrive to infuse into his brain—they cunningly introduce some tame snakes into some crevice of his house, which come to their master as soon as he sounds his musical call. The chuckling enchanter then instantly whips up the serpent, claps it into his basket, pockets his fee, and, all the while doubtless laughing in his sleeve, goes to some other house, to renew his offers of assistance to similar dupes.”

As to the idea that the snakes are previously deprived of their fangs, and that the jugglers secure themselves against all danger of being injured by the regular dancing snakes that they carry about with them in baskets, a single anecdote related by Forbes, in his *Oriental Memoirs*, will I think suffice to combat and refute it. Not having the book by me while I write, I hope my readers will excuse any slight discrepancies which they may detect on a reference to my authority. Forbes states that on the cessation of the music the reptiles lapse into a sort of lethargy, and appear motionless. It is, however, he adds, necessary that they should be immediately covered up in the baskets, as otherwise they may spring upon and wound the spectators; and he informs us that fatal accidents frequently occur from inattention to this precaution. Amongst his drawings is that of a Cobra de Capella, which, under the magic influence of a professed serpent-charmer’s music, danced before him for an hour upon his table while he painted it, and during that period he repeatedly handled it and carefully examined the structure of its head, hood, and jaws, and inspected minutely the variety and extreme beauty of its spots. The following day an upper servant of his rushed into his apartment, and cried out that he was a fortunate, a most fortunate man, doubtless under the immediate protection of the Prophet—that his devotions had proved acceptable, and sundry other expressions, totally incomprehensible to Forbes, who inquired his meaning. The man then related that he had just been in the bazaar, where he had seen the same juggler who had entertained him the day preceding, performing before a crowd of people, who, as was usual on such occasions, formed a circle around the operator, seated on the ground. At the close of the performance, the reptile, whether infuriated from the music ceasing too suddenly, or from some other cause not to be explained, darted amongst the spectators, and seizing a young woman by the throat, inflicted a wound of which she died in about an hour. Here was proof positive that the extraction of the serpent’s fangs was thought by no means essential to training him to his performance.

So much for the idea that the *dancing* snakes are always deprived of their fangs—now as to the reality of the circumstance of the *wild* serpents being drawn forth from their holes by the charmer’s pipe, and not being *tamed animals* placed in those holes for the express purpose of deception.

Perhaps the best refutation of this idea that I can adduce, will be found in a highly interesting account I received lately from a friend resident for many years in India, and who directed a more than ordinary degree of attention to snake-charmers and their feats; nay, not merely to them, but to every other description of magical rites, of which no land now furnishes so many wonder-working adepts as India, not even Egypt.

He told me of men who would sow a seed of corn in a flower-pot, and by sundry mysterious incantations cause it to sprout, grow up, throw off leaves, bud, produce grain, and ripen, all within the space of an hour. He told me of men who would turn an empty hamper upside down, and produce from thence shawls, jewels, strings of beads, muslin turbans, and, in short, any article the spectators chose to demand. He told me many other singular and wondrous stories; but, what at present is of more immediate importance, he gave me a singular account of serpent-charming. I need not recapitulate its details, as they precisely resemble those quoted in a former article: I need

only observe, that he assured me he had examined the subject too closely, and had taken too many precautions to prevent the possibility of fraud, to admit of its being, in any one instance, practised upon him. He had sent a distance of fifty miles up the country for a snake-catcher, and had set him to work in a spot entirely unknown to all as the place he had selected, until he conducted them and the juggler thither; and he had dozens of times seen the reptiles drawn from their retreats by the sounds of the flute or fife, which they evidently derived extreme pleasure from hearing. It was my friend's opinion that the chief agent in the operation of serpent-charming was music; the animals positively delighted in the sound of the soft instruments employed by the performers, and were by its influence lulled into a sort of pleasurable trance whenever the exciting cause was put in operation.

My friend once sat beneath the shade of a spreading tree, and was amusing himself with his flageolet, an instrument on which he performed with much skill; he had not been thus employed above an hour, when a native, happening to come up the approach to his residence, suddenly started, and began muttering prayers as fast as he was able. My friend could scarcely refrain from laughing at this singular exhibition, being entirely ignorant of its cause, and was about to rise up, when the stranger called out to him to remain where he was, and keep playing upon his instrument if he valued his life, for that imminent danger threatened him. This announcement, instead of producing the desired effect, only confirmed my friend in the supposition that the strange Hindoo was some mad fakir, who, half knave and half crazy, was endeavouring to play upon his feelings, as he so frequently and successfully did upon those of his silly countrymen. He accordingly sprang to his feet; but what his consternation was, you, reader, may judge. As he rose, a prodigious Cobra de Capella presented itself to his astonished and affrighted gaze, hanging by its tail from the tree, its gleaming eyes and hooded head not more than two feet from his own! For a moment he felt as it were fascinated, rooted to the spot; but in a second afterwards, terror acted in her more legitimate manner: he sprang several paces backward, and running to the house, procured assistance, on which he again sallied forth, accompanied by several natives, who by their cries and hooting succeeded in inducing the snake to beat a retreat. He was watched, however, in his departure, and traced to a hole; a guard was placed over it, and that too of Europeans, so that no confederacy could exist. A snake-catcher was procured from a distance of ten miles; he approached the hole, played upon his instrument, and at length the reptile crawled forth, and was captured and secured in the usual manner.

I think that even this brief and hurried account must have compelled my readers to cast from their minds all notion of the snakes being *laid in the proper places* by the jugglers beforehand, as preparatory to a performance, as I have shown in the instances above mentioned that no such thing could have been done. And the idea of the creature's having been previously rendered harmless, is also overturned by the circumstance of the Cobra de Capella, handled one day with impunity by Forbes, having on the following morning bitten a young woman, who died of the effects of the poison within an hour. I trust, then, that I have brought you to admit that the art of snake-charming is a *genuine* art, whether simple or not remains to be proved when the true secret shall have been found out; and that the professors of this secret are not impostors, at least not in this particular, but at the very least as respectable characters as the rat-catchers of our native country, who, my readers are of course aware, pretend likewise to possess the secret of charming and enticing rats from any place. In my next paper I shall conclude this subject of *charming*, and endeavour to explain some of the modes by which various animals are thus seduced.

H. D. R.

KNOWLEDGE AND IGNORANCE.—No. I.

BOULDERS.

In using the above terms, let it not be supposed that I mean to imply by the one a perfect knowledge, or a knowledge of everything, and by the other a perfect ignorance, or a total want of any knowledge. Either of such conditions of the mind is incompatible with human organization; the one, a perfect knowledge, belongs alone to an order of intelligence infinitely excelling that of man; and the other, a perfect ignorance, must be sought for in creatures so far below him as to possess no intelligence. The idiot is not without perception and knowledge, though of an imperfect and irregular kind. The dog knows its master, recognizes and obeys his voice. The horse knows and traces, after years of absence, the road he had once been accustomed to travel; and even reptiles and fishes acquire a knowledge of persons, of times, and of things; all this being independent of that range of intelligences which has been given to every creature for the preservation of its own existence, and for ensuring the continuance of its species. The terms Knowledge and Ignorance are used, then, in a comparative sense, being, according to circumstances, convertible one into the other. What, for instance, is knowledge at one time, becomes ignorance at another; and the man who seems wise to those who know less than he does, seems equally foolish to those who know more—a strong reason surely why no one, however gifted he may to himself appear, should despise his less gifted brethren. Mounted he may indeed be on a hill so high that he can discern objects in the distance which are hidden from the more humble plodders of the plain below, and yet his own horizon be proportionately limited when compared to that of others who have climbed the still higher mountain above him. Can we not all bring home to our minds this varying value of our acquirements at successive periods of our lives? and are we not sometimes surprised to reflect that some problem was once difficult, or some fact obscure, which is now as familiar to our understandings as the daylight to our eyes? We have, in short, as regards these particular objects, passed from the night of ignorance into the day of knowledge. And us with the same individual, and even with whole classes of individuals, at different epochs, so is it with different individuals at the same time: one person holding in his hand the dim taper of ignorance, sees by its flickering and ill-directed light the object of his examination, distorted by partial and shifting shadows—just as some timid traveller on a dusky night sees in each waving bush, as to his alarmed imagination it grows to a portentous size, or assumes a fearful form, some aërial phantom, or some terrestrial monster. The other, raising the bright lamp of knowledge, dispels at once by its clear and steady light, uncertainty, and sees the object as it is.

So many indeed are the practical illustrations of the different manner in which the same object is viewed by knowledge and by ignorance, that it is difficult to make a first choice. All around us there are objects, the nature and qualities of which are known to the few, unknown to the many, and hence either overlooked or misunderstood by the latter, studied and understood by the former. Each portion, however minute, of our own body, and of that of every other organic being, has in it wherewithal to exercise the ingenuity and reflection of the wisest; and yet how many thousands

live and die without having even desired much less sought after such knowledge! Nor is the inorganic world less fruitful in subjects of inquiry, nor less neglected. The ploughman “whistles as he goes for want of thought,” not because nature has failed to spread around him inexhaustible food for thought, but because his mind has not been trained to think. By each movement of his ploughshare, page after page, as it were, is opened to his view of new and interesting matter—and yet he sees before him nothing but silent and unmeaning clods. By each movement of his foot he disturbs those pebbles which, speechless to him because he questions not, return to the interrogations of knowledge wonder-stirring answers, when asked,

1. Of what they are composed?
2. Whence they came?
3. And how they came?

For the present we shall pass over these more humble whisperers of things curious and strange, and turn to those massive fragments of rocks which, far removed from their original site, are now scattered either singly or in groups over a large portion of the earth’s surface, resting sometimes on the slopes of hills composed of materials totally different from their own, seen sometimes on the sand and gravel of extensive plains, and distant from the mountains of which they were once a part, sometimes from one to three hundred miles: they are Boulders. Can we not picture to ourselves, in that remote period of our island’s history when forest and morass occupied the place of its bogs, and when the winds sighed over comparative desolation, an ancient inhabitant, imbued with nature’s living poetry, pausing before one of those grey lichen-covered masses which had withstood the warrings of the elements for perhaps thousands of years, and, as the awe of the surrounding solitude came like a charm over his soul, gazing with growing veneration at the venerable rock?—to him it would appear as if cast down from heaven, or planted where it now stands by some supernatural or giant hand. What spot, then, more fitted for the simple worship of nature’s child?—what temple, what altar more suited to his simple rites?

A rock such as we have here described may have been found supported in part by lesser fragments, or such supports may have been introduced by partial excavations under favourable projections of its surface; and in either case, the superfluous earth, sand, or stones under and about it, being removed, this ancient monument of the operations of Nature would henceforth become an instrument in the worship of Nature’s God—a Cromlech!

Whether, however, this be, or not, a correct view of the original impulse which led to the selection of these giant stones, or of the purpose to which they were applied, it is for our antiquarian friends to decide. Suffice it here to add, that the transportation of such huge masses from their native beds, by the power of man or of giants, was at such a remote epoch, and under the circumstances of the country, impossible; nor will I stop to inquire whether a work so mighty was performed by spirits light as air.

Let us turn to the consideration of the phenomenon of Boulders, as it has appeared to the eye of science. And perhaps there are no two facts which place it in so strong a light, and embrace so fully the reasonings founded upon it, as the dispersion of blocks of the granite and other rocks of Sweden over a large portion of Northern Europe, the boulders, either singly or in clusters, being disposed in long parallel lines or trainées, for upwards of two hundred miles from the mountains of Scandinavia, to which, by identity of mineral composition, they have been traced, although separated from them by the Baltic Sea; and the occurrence of boulders of alpine granite resting on the secondary rocks of the Jura chain, between which and the Alps are situated the deep valley of the Rhone, the Lakes of Geneva and Neufchatel, the distance travelled by the boulders being sixty

miles. Saussure, struck by the spectacle of clusters of these fragments so far removed from any rock resembling them, declared that they looked as if rained down from heaven; a sentence strikingly expressive of the difficulties which attend on an explanation of their occurrence. De Luc rightly speaks of such travelled masses of stone as being “one of the most important of geological monuments, since they offer a rigorous criterion of the different systems concerning the revolutions which have happened on our globe;” and in describing the vicinity of Cuxhaven, situated at the extremity of the Bremen country, which lying between the Gulfs of the Elbe and Weser, is as it were a peninsula, he cites the very forcible example it affords of a vast abundance of boulders at a distance of more than two hundred miles from the Scandinavian chain, the outlet, itself sixty miles wide, of the Baltic, forming part of the intervening space.

At the time of De Luc’s visit to Cuxhaven (1797), a dike was constructing to secure the port from the violence of the sea, and the plan of employing blocks for this purpose was suggested by the quantity which were scattered over all the neighbouring country. From the vicinity alone of Hornburg, an inland town between the ports of Stade and Harborg, 600 lasts of blocks, amounting to 240,000 quintals, or 23,679 tons, had at that time been brought and consumed in the dike, which, with the thickness necessary to resist the utmost impetuosity of the waves, and a height of about eight feet, already extended three leagues to the westward of the town. The country in which these accumulations of erratic boulders had taken place, is an expanse of sand covered with heath, except where broken by cultivated patches around the scattered villages, the surface being undulated by hills composed either of sand or of heaps of boulders. De Luc adds, “that he travelled ten miles without perceiving in the whole horizon any house, or even a hovel, or a single tree”—desolate and dreary indeed to the eye of painter or poet, yet rich in all the elements of sublimity to the eye of the geologist.

It is quite unnecessary to adduce other and less imposing examples from Great Britain and Ireland of similar facts, the difficulties of explanation being fully embraced by those selected. How have they been brought to their present places? is then the question mentally asked, as well by the learned as the unlearned.

Saussure, celebrated for his examination of the Alps, imagined a great debacle and retreat of the sea from the strata that had been formed, as he supposed, by chemical precipitations; and to the violent rush of the vast current he ascribed the excavation of the valleys, and the transport of immense masses of stones from the central chain of the Alps, beyond the precincts of those mountains, to the Jura. Here, then, the excavation of the valleys of the Alps, and the transport of the boulders, are considered results of one great catastrophe, by which the bottom of the sea became hard dry land, its waters descending into huge abysses which had burst open around the Alps. The phenomenon of Boulders is general in a large portion of the northern hemisphere; the explanation however is local and hence insufficient; whilst the philosopher’s machinery, of huge abysses, like the peasant’s giant, is born of necessity, not deduced from experience.

Others, and even yet they are many, attribute the transport of both gravel and boulders to the Noachian deluge, which is their great geological catastrophe. The application, however, of that great historical event to such physical agencies, is beset with great difficulties. The words of scripture do not support, but rather oppose, the notion of a huge wave rising in the north to a great height, then rushing southwards over the dry land, and rooting up or sweeping before it, by hydrostatic pressure, fragments of the earth’s crust. Nor are facts more in accordance with that notion—the boulders of Scandinavia were moved from north to south—the boulders of the Alps from south to north, passing over the Jura mountains into Franchcomté—the stratification of many of the heaps of sand and gravel—the position of the boulders generally on the surface, whether of

rocks, of sand, or of gravel—and the valleys, lakes, and seas now lying in the line of movement, which, if existing before the catastrophe, must have been filled up before the boulders could have travelled farther, if formed after, must have required the action of a second catastrophe of equal violence for their formation. And if, which is more in accordance with scripture, we consider the waters rising from the surrounding seas over the dry land, and then suppose them urged on with immense velocity, the effect would be a heaving up and moving forward of fragments from the lower land, by which the surface of the higher would be partly covered and protected; and at the return of the waters to their ancient beds, these fragments would be swept off, and carried back the same way they came. Neither, then, the words of scripture, nor the facts themselves, require us to seek in the Noachian deluge for an explanation of these phenomena. Another theory, still adhered to by many modern geologists, is, the action of submarine currents, at a time when the present dry land had only in part emerged from the sea. This theory has the advantage of dealing with bodies of diminished gravity, in consequence of their immersion in a fluid, and consequently of having to provide for the movement of weights less by one-half or one-third than they would have been in air. In conjunction with the theory of raised beaches, it explains many of the phenomena of accumulations of sand and gravel, but not all. And as regards the transport of boulders, it fails; the great size and angular form of some—their occurrence at various levels, resting on various strata—sometimes connected with, and sometimes unconnected with sand or gravel—their position frequently on the top of heaps and ridges of gravel, being facts in seeming opposition to such an explanation, even were it conceded that all the depressions now existing on the line of travel, as lakes and seas and valleys, were scooped out subsequently to their transport.

The geological system of the illustrious Hutton assumed as an essential principle, that as the present continents and dry land were once the bottom of the ocean, and have been formed, either in greater part or entirely, of fragments of pre-existing continents now submerged, so is the work of destruction and renewal still continuing, the substance of our present dry land being loosened, abraded, or worn down by meteoric agencies, and carried by torrents and rivers to the ocean, to be there by currents distributed over the bottom of the sea, and by internal heat consolidated into new strata, which in time will be elevated into new continents and islands. To apply this theory in the case of the Jura boulders, Playfair assigned their transport to an epoch anterior to the formation or excavation of the deep valleys and lakes which would now form an insurmountable obstacle to such transport, and thus obtained a greatly inclined plane, extending from the summit of the Alps to the Jura, on which to trundle the fragments gradually downwards, by aid of the numerous streams and torrents descending from the higher to the lower ground. But as this theory would, as thus applied, premise that the land had been raised above the sea-level prior to the transport of the boulders, no means of effecting the great excavations, including the Lakes of Geneva and Neufchatel, which are supposed to have been formed subsequently, are left, except the slow erosive action of rains, frost, torrents, and such-like agents—means which few will consider adequate to the desired object; and hence the explanation of Playfair, resting solely on a bold hypothesis, must be rejected. As most of the preceding theories referred to the usually rounded condition of the granite boulders (many boulders of other rocks are angular), as an evidence of movement through the agency of water, De Luc, preparatory to the promulgation of his own theory, thought it expedient to show that blocks of granite, even as they stand tranquilly braving the storms, are gradually weathered into a rounded form. He thus cites the granite of Darmstadt as an example:—"Here I found a striking example of the manner in which blocks and even rocks of granite are rounded by the decomposition of the angles of their masses. I perceived it first in some angular pieces that had been detached and lay at the foot of the rock, surrounded with rubbish; for, on giving them a strong blow with an iron at the end of my stick, the angles fell off, detaching themselves with a concave surface on their inner side; and I thus produced rounded blocks, exactly

resembling those which I had seen scattered on the plains.” This spherical concretionary structure has been noticed in the granite of Dublin and Down, and is common in trap rocks. Having smoothed away this difficulty, De Luc tacks on the boulders as a corollary to his theory of subsidences. Immense masses of strata, subsiding into huge caverns or hollows beneath them, fragments of the lower strata were broken off and blown upwards by the force of the pent-up air and gases rushing through the cracks of the sinking strata, the weight of which continued more and more to compress them, so that the boulders of M. De Luc came from below, and not from above. This is also a gratuitous hypothesis; and as the localities of many boulders exhibit no signs of such subsidences and explosions, it has obtained few if any adherents. So far, then, it would appear that philosophers, though armed with all the powers of mind invigorated by study and sharpened by research, have fought in vain against the difficulties which like a rampart fence in this rugged problem. For a moment they have appeared illumined by the light of knowledge, and have then sunk into the darkness of ignorance. But though philosophy may yield, she never will despair. And now, having marshalled new forces for the combat, we shall see her, with brighter hopes and prospects, again renew the assault. To the consideration, therefore, of a widely different class of explanations, I shall proceed to direct attention in a second paper.

J. E. P.

INTELLECTUALITY OF ANIMALS.—Father Bougeant, a Jesuit, was placed in confinement by his superior in the College of La Fleche, near Paris, for what he had written on the subject of the intellectuality of animals. His views, if not orthodox, were certainly curious and amusing, and there is a sprightliness in his mode of treating the subject, graceful at least in the Frenchman, if not conformable to the divine. The following observations, extracted from that section of his work which treats of the language of beasts, may amuse the reader:—“Our first observation upon the language of beasts is, that it does not extend beyond the necessities of life. However, let us not impose upon ourselves with regard to this point. To take things right, the language of beasts appears so limited to us only with relation to our own; however, it is sufficient to beasts, and more would be of no service to them. Were it not to be wished that ours, at least in some respects, were limited too? If beasts should hear us converse, prate, lie, slander, and rave, would they have cause to envy us the use we make of speech? They have not our privileges, but in recompense they have not our failings. Birds sing, they say; but this is a mistake. Birds do not sing, but speak. What we take for singing is no more than their natural language. Do the magpie, the jay, the raven, the owl, and the duck, sing? What makes us believe that they sing is their beautiful voice. Thus, the Hottentots in Africa seem to cluck like turkey-cocks, though it be the natural accent of their language; and thus several nations seem to us to sing, when they indeed speak. Birds, if you will, sing in the same sense, but they sing not for singing’s sake, as we fancy they do. Their singing is always an intended speech; and it is comical enough that there should be thus in the world so numerous a nation which never speak otherwise but tunably and musically. But, in short, what do those birds say? The question should be proposed to Apollonius Tyaneus, who boasted of understanding their language. As for me, who am no diviner, I can give you no more than probable conjectures. Let us take for our example the magpie, which is so great a chatterer. It is easy to perceive that her discourses or songs are varied. She lowers or raises her voice, hastens or protracts the measure, lengthens or shortens her chit-chat; and these evidently are so many different sentences. Now, following the rule I have laid down, that the knowledge, desires, wants, and of course the expressions of beasts, are confined to what is useful or necessary for their preservation,

methinks nothing is more easy than at first, and in general, to understand the meaning of these different phrases.”—*Dublin University Magazine*.

ATMOSPHERIC RESISTANCE ON RAILWAYS.—In Dr Lardner’s third lecture on railways at Manchester, he detailed a variety of experiments made in order to ascertain the source of resistance. “He found that an enlarged temporary frontage constructed with boards, of probably double the magnitude of the ordinary front of the train, caused an increase of resistance so trifling and insignificant as to be entirely unworthy of account in practice. Seeing that the source of resistance, so far as the air was concerned, was not to be ascribed to the form or magnitude of the front, it next occurred to him to inquire whether it might not arise from the general magnitude of the train front ends, top and all. An experiment was made to test this. A train of waggons was prepared with temporary sides and ends, so as to represent, for all practical purposes, a train of carriages, which was moved from the summit of a series of inclined planes, by gravity, till it was brought to rest; it was next moved down with the high sides and ends laid flat on the platform of the waggons, and the result was very remarkable. The whole frontage of the latter, including the wheels and every thing, a complete transverse section of the waggons, measured 24 feet square, and with the sides and ends up, so as to present a cross section, it amounted to nearly 48 square feet. The uniform velocity attained on a plane of 1 in 177, without the sides up, was nearly 23 miles an hour; whereas, with the sides up, it was only 17 miles an hour; so that, as the resistance would be in proportion to the square of the velocity, other things being the same, there would be a very considerable difference, due to that difference of velocity. Then, at the foot of the second plane, while the sides were down, an undiminished velocity remained of $19\frac{1}{2}$ miles an hour, whereas, with the sides up, it was reduced to $8\frac{1}{2}$ miles an hour; so that a very extensive difference was produced. They would see at once that this was a very decisive experiment to prove that the great source of resistance was to be found in the bulk, and not the mere section or the form, whether of the front or the back of a train; but simply in the general bulk of the body carried through the air. It was very likely to arise from the successive displacements of a quantity of the atmosphere equal to the bulk of the body; or still more probably, from the fact of the extensive sides of the train; and indeed there was little doubt that the magnitude of the sides had a very material influence; for if they consider what is going on in the body of air extending from either side of a train of coaches, they would soon see what a mechanical power must be exercised upon it. Thus, when a train is moving rapidly, the moving power had not only to pull the train on, but it had to drag a succession of columns of air, at different velocities, one outside the other, to a considerable extent outside the train; and it did more, for it overcame their friction one upon the other; for as these columns of air were at different velocities, the one would be rubbing against the other; and all this the moving power had to encounter. This would go far to explain the great magnitude of resistance found, and its entire discordance with any thing previously suspected.”

GILDING OF METALS BY ELECTRO-CHEMICAL ACTION.—M. de la Rive has succeeded in gilding metals by means of this powerful action. His method is as follows: he pours a solution of chloride of gold (obtained by dissolving gold in a mixture of nitric and muriatic acid) as neutral as possible and very dilute, into a cylindrical bag made of bladder; he then plunges the bag into a glass vessel containing very slightly acidulated water; the metal to be gilded is immersed in the solution of

gold, and communicates by means of metallic wire with a plate of zinc, which is placed in the acidulated water. The process may be varied, if the operator pleases, by placing the acidulated water and zinc in the bag, and the solution of gold with the metal to be gilded in the glass vessel. In the course of about a minute, the metal may be withdrawn, and wiped with a piece of linen; when rubbed briskly with the cloth, it will be found to be slightly gilded. After two or three similar immersions the gilding will be sufficiently thick to enable the operator to terminate the process.—*Athenæum*.—[By referring to the article on the Electrotpe which appeared in No. 20 of the Irish Penny Journal, the reader will be enabled clearly to understand the mode in which the gold is separated from the acid, which holds it in solution, and forced, or attracted, to deposit its particles upon the metallic surface; the solution of gold bearing in this case a precisely similar relation to the metal plate, as the solution of copper in the other.]

DEFINITION OF CHERUB.—A lady (married of course) was once troubled with a squalling brat, whom she always addressed as “my cherub.” Upon being asked why she gave it that appellation, she replied—“Because that it is derived from cherubim, and the Bible says, ‘the cherubims continually do cry.’”

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